US Flight Delays 2015

Advanced findings and key technical work from the Power BI project

The dataset covers 5,819,079 US domestic flights in 2015, across 14 airlines and 322 airports. A flight counts as delayed when it arrives 15 or more minutes late. The most interesting findings are the ones that go against the obvious reading of the data. The hour-of-day, correlation and recovery results came from direct queries on the model and are not visuals in the report.

Advanced findings

Finding	Evidence	Why it's interesting
Early flights are rarely late, but late ones run long	5 AM flights: 7.3% delayed, averaging 72.2 min when late. 10 PM flights: 22.9% delayed, averaging 52.2 min.	How often delays happen and how long they last are separate measures, so one KPI can mislead.
A low delay rate can hide a bad day	January 27 had a 7.1% delay rate (the daily minimum is 6.4%), but 2,884 flights were cancelled, about 19% of that day's flights.	Cancelled flights have no arrival delay, so they drop out of the rate. The two metrics should be read together.
Some delays start after departure	232,701 late arrivals (22%) left less than 15 minutes late.	Departure performance alone misses about one late arrival in five.
Most flights arrive early	61% of flights with arrival data landed ahead of schedule. Average delay is 9.4 min at departure but 4.4 min at arrival.	This could reflect time recovered in the air, generous schedules, or both; the data can't separate them.
Airlines have different delay causes	Southwest: late aircraft are about 51% of the main cause minutes. Delta: airline-caused minutes (2.71M) exceed late aircraft (2.14M). Spirit: air system is largest (0.94M).	Each airline would need a different fix, which a single ranking hides.
Evening risk is about 2.4 times morning risk	5-8 AM departures: 10.7% delayed. 5-8 PM departures: 25.6%.	This fits delays building through the day, though it doesn't prove that's the cause.

Key technical work

1. DAX ranking that holds up in any table

The measure behind the Top 15 Airports visuals. Ranking over the whole `airports` table stops city and state columns from narrowing the ranking.

```
Top 15 Origin Delayed =
IF(
    NOT ISBLANK(SELECTEDVALUE(airports[AIRPORT]))
        && RANKX(
            FILTER(ALLSELECTED(airports), NOT ISBLANK(airports[AIRPORT])),
            [Delayed Flights]
        ) <= 15,
    [Delayed Flights]
)
```

2. Using an inactive relationship for destination airports

A model can have only one active relationship between `flights` and `airports`, so destination measures switch to the inactive one.

```
Destination Delayed Flights =
CALCULATE(
    [Delayed Flights],
    USERELATIONSHIP(flights[DESTINATION_AIRPORT], airports[IATA_CODE])
)
```

3. Date table that follows the data

The start and end dates come from the flights themselves.

```
VAR StartDate = MIN(flights[FLIGHT_DATE])
VAR EndDate = MAX(flights[FLIGHT_DATE])
RETURN ADDCOLUMNS(CALENDAR(StartDate, EndDate), "Year", YEAR([Date]), ...)
```

4. One chart for several columns

A small `Delay Cause` table and a `SWITCH` measure let five separate cause columns share one sorted bar chart.

```
Delay Minutes by Cause =
SWITCH(
    SELECTEDVALUE('Delay Cause'[Cause]),
    "Late aircraft", SUM(flights[LATE_AIRCRAFT_DELAY]),
    "Airline", SUM(flights[AIRLINE_DELAY]),
    ...
)
```

5. Pearson correlation written directly in DAX

This query produced the r = 0.28 between daily flight volume and delay rate, with no Python or R involved.

```
DEFINE
    VAR T = SUMMARIZECOLUMNS('Date'[Date], "F", [Total Flights] + 0, "R", [Delay Rate] + 0)
    VAR mF = AVERAGEX(T, [F])
    VAR mR = AVERAGEX(T, [R])
EVALUATE
ROW("corr",
    SUMX(T, ([F] - mF) * ([R] - mR))
        / SQRT(SUMX(T, ([F] - mF) ^ 2) * SUMX(T, ([R] - mR) ^ 2)))
```

6. Testing ideas without changing the model

`DEFINE COLUMN` creates a column for one query only, which is how the hour-of-day analysis ran on 5.8 million rows.

```
DEFINE COLUMN flights[H] = INT(VALUE(flights[SCHEDULED_DEPARTURE]) / 100)
EVALUATE
SUMMARIZECOLUMNS(flights[H], "Rate", [Delay Rate], "Avg", [Avg Delay When Late])
```

7. Report and model built as code

Technique	Used for
Power BI project format (PBIP) and TMDL	Storing the model as text files, including tables, measures and relationships
Tabular Object Model (TOM)	Creating the date table and relationships in the running model
ADOMD.NET and DAX queries	Testing measures and checking results against the source
Power Query (M)	Loading the CSVs with explicit column types, keeping values like 0005 as text
Python	Generating the report.json pages, plus a scan of flights.csv to profile types, blanks and unmatched keys
Field-level file comparison	Checking that Desktop's saves contained no manual edits before overwriting them

Limits

- **Delay causes:** the data records cause minutes, but it can't prove why a pattern happens. The links between delays through the day are a reasonable reading, not a tested cause.
- **Scripted build:** editing report.json directly isn't an officially supported way to build reports. It suits repeatable builds, but the Power BI interface is safer for everyday edits.
- **Scope:** this is one year of data, so patterns can't be called seasonal.